

UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
GCE Advanced Subsidiary Level and GCE Advanced Level

MARK SCHEME for the May/June 2012 question paper
for the guidance of teachers

9709 MATHEMATICS

9709/61

Paper 6, maximum raw mark 50

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes must be read in conjunction with the question papers and the report on the examination.

- Cambridge will not enter into discussions or correspondence in connection with these mark schemes.

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Mark Scheme Notes

Marks are of the following three types:

M Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

B Mark for a correct result or statement independent of method marks.

- When a part of a question has two or more “method” steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly when there are several B marks allocated. The notation DM or DB (or dep*) is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.
- The symbol $\surd$ implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only. A and B marks are not given for fortuitously “correct” answers or results obtained from incorrect working.
- Note: B2 or A2 means that the candidate can earn 2 or 0.
B2/1/0 means that the candidate can earn anything from 0 to 2.

The marks indicated in the scheme may not be subdivided. If there is genuine doubt whether a candidate has earned a mark, allow the candidate the benefit of the doubt. Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored.

- Wrong or missing units in an answer should not lead to the loss of a mark unless the scheme specifically indicates otherwise.
- For a numerical answer, allow the A or B mark if a value is obtained which is correct to 3 s.f., or which would be correct to 3 s.f. if rounded (1 d.p. in the case of an angle). As stated above, an A or B mark is not given if a correct numerical answer arises fortuitously from incorrect working. For Mechanics questions, allow A or B marks for correct answers which arise from taking g equal to 9.8 or 9.81 instead of 10.

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The following abbreviations may be used in a mark scheme or used on the scripts:

AEF	Any Equivalent Form (of answer is equally acceptable)
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
BOD	Benefit of Doubt (allowed when the validity of a solution may not be absolutely clear)
CAO	Correct Answer Only (emphasising that no “follow through” from a previous error is allowed)
CWO	Correct Working Only – often written by a ‘fortuitous’ answer
ISW	Ignore Subsequent Working
MR	Misread
PA	Premature Approximation (resulting in basically correct work that is insufficiently accurate)
SOS	See Other Solution (the candidate makes a better attempt at the same question)
SR	Special Ruling (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)

Penalties

- MR –1 A penalty of MR –1 is deducted from A or B marks when the data of a question or part question are genuinely misread and the object and difficulty of the question remain unaltered. In this case all A and B marks then become “follow through ✓” marks. MR is not applied when the candidate misreads his own figures – this is regarded as an error in accuracy. An MR –2 penalty may be applied in particular cases if agreed at the coordination meeting.
- PA –1 This is deducted from A or B marks in the case of premature approximation. The PA –1 penalty is usually discussed at the meeting.

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1 $z_1 = \frac{30-28.3}{\sqrt{4.5}} = 0.8014$ $z_2 = \frac{25-28.3}{\sqrt{4.5}} = -1.5556$ $\Phi_1 - (1 - \Phi_2) = 0.7884 + 0.9401 - 1$ $= 0.729$	M1	Standardising at least one value, sq rt.ess; no cc
	M1	$\Phi_1 + \Phi_2 - 1$ oe
	A1 [3]	Correct answer
2 (i) $0.25p = 0.075$ $p = 0.075/0.25 = 0.3$ (ii) $P(2 M) = \frac{P(2 \text{ and } M)}{P(M)}$ $= \frac{0.45 \times 0.85}{0.3 \times 0.1 + 0.45 \times 0.85 + 0.25 \times 0.3}$ $= \frac{0.3825}{0.4875}$ $= 0.785$	B1 [1]	Answer given, must show some working
	M1	attempt at cond prob with single prod in num and Σ three 2-factor o.e prods in denom
	B1	correct numerator of a fraction
	A1	correct unsimplified denom
	A1 [4]	correct answer
3 (i) $p = 0.1$ (ii) (a) $P(X=1, Y=3) = 0.3 \times 0.2 = 0.06$ $P(X=2, Y=2) = 0.15 \times 0.5 = 0.075$ $P(X=3, Y=1) = 0.3 \times 0.3 = 0.09$ $P(\text{sum is } 4) = 0.225$ (b) $P(X=1, Y=\text{anything}) = 0.3$ $P(X=2, Y=\text{anything}) = 0.15$ $P(X=3, Y=1, 2) = 0.3 \times 0.8 = 0.24$ $P(X=4, Y=1) = 0.2 \times 0.3 = 0.06$ $P(X=5, Y=1) = 0.05 \times 0.3 = 0.015$ $P(\text{product} < 8) = 0.765$ OR $P(Y=1, X=\text{anything}) = 0.3$ $P(Y=2, X=1, 2, 3) = 0.5 \times 0.75 = 0.375$ $P(Y=3, X=1, 2) = 0.2 \times 0.45 = 0.09$ $P(\text{product} < 8) = 0.765$	B1 [1]	
	M1	Summing 2 or 3 options
	B1	One option correct unsimplified
	A1 [3]	correct final answer
	M1	Σ 3 or more two-factor options
	B1	Two correct options
	A1 [3]	Correct answer
	M1	
	B1	
	A1	
4 (i) $P(X < 5) = 1 - P(5, 6, 7)$ $= 1 - (0.21)^5(0.79)^2 {}_7C_5 - (0.21)^6(0.79) {}_7C_6 - (0.21)^7$ $= 0.994$ (ii) $P(\text{at least } 1) = 1 - P(0) = 1 - (0.79)^7$ $= 0.808$ $P(\text{exactly } 3 \text{ weeks}) = (0.808)^3(0.192) {}_4C_3$ $= 0.405$	M1	Binomial expression with powers $\Sigma 7$ and probs $\Sigma = 1$, and ${}_7C_r$
	A1	Correct unsimplified expression
	A1 [3]	Correct answer
	M1	Attempt to find $P(\text{at least } 1)$ or $1 - P(0 \text{ and } 1)$
	A1	Rounding to correct answer
	M1	Bin expression with powers $\Sigma 4$ and their 0.808 etc. and ${}_4C_3$
	A1 [4]	Correct answer

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5 (i) Flat screen $\begin{array}{r} 6 \\ 9 \ 5 \\ 6 \ 4 \ 2 \ 1 \\ 7 \ 4 \end{array}$ conventional $\begin{array}{r} 6 \\ 7 \\ 8 \\ 9 \\ 10 \end{array}$ $\begin{array}{r} 5 \ 7 \ 9 \\ 1 \ 4 \ 5 \ 7 \\ 5 \ 6 \end{array}$ key 5 8 4 means 0.85 m for flat screen and 0.84 m for conventional (ii) Conventional median = 0.74 conv IQ range = 0.81 – 0.68 = 0.13 (iii) mean = 0.927 sd = 0.0882	B1	Correct stem must be integers
	B1	Correct flat screen leaves
	B1	Correct conventional screen leaves
	B1 [4]	Key must have units and TV type
	B1	Correct median
	M1	Their UQ – their LQ
	A1 [3]	Correct answer
	B1	
	B1 [2]	Need 3 s.f. (Accept 0.0878 to 0.0889)
6 (i) $-1.253 = \frac{6 - \mu}{\sigma}$ $0.648 = \frac{12 - \mu}{\sigma}$ $\mu = 9.9$ $\sigma = 3.15$ or 3.16 (ii) need $P(z < -1 \text{ or } z > 1)$ $= 1 - \Phi(1) + \Phi(-1)$ $= 2 - 2 \times 0.8413$ $= 0.3174$ number = 317	B1	$Z = \pm 1.253$
	B1	$Z = \pm 0.648$
	M1	Any equation with μ and σ and a reasonable z value not a prob. Allow cc or –, not $\sqrt{\sigma}$ or σ^2
	M1 [Indpt]	Att. to solve by substitution or elimination
	A1 [5]	
	B1	$z = 1$ or -1 seen
	M1	Correct area i.e. $2 - 2\Phi$
	M1	Mult their prob if sensible, by 1000
	A1 [4]	Accept 317, 317.4, 318
7 (a) (i) 7 couples in 7! ways each couple in 2 ways so $7! \times 2^7$ $= 645120$ OR $14 \times 12 \times 10 \times 8 \times 6 \times 4 \times 2 =$ 645120 (ii) $7! \times 7! \times 2$ $= 50,803,200$ (50,800,000) OR $14 \times 6! \times 7!$	B1	7! seen multiplied
	M1	mult by 2^7
	A1 [3]	correct final answer
	B2	correct unsimplified answer
	A1	correct answer
	B1	$7! \times 7!$ seen
	B1 [2]	Correct answer
	B1	$14 \times 7!$ seen
	B1	Correct answer

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(b) (i) $7C2 = 21$ (ii) all in: 1 all not in: $5C4 = 5$ total 6 (iii) 2 girls in: $6C2 \times 3C2 = 45$ 3 girls in: $6C1 = 6$ Total 51	B1	[1]	
	M1		Considering both cases
	A1	[2]	Correct answer
	M1		Attempt at summing 2 and 3 girls in the team need not see $3C2$
	A1	[2]	Correct answer